Q) E.M.F of an electrical cell is 2 volt. A 10 Ω resistance is joined at its two ends then potential difference is measured 1.6 volt. Find out the internal resistance and lost volt. (1+1)

Ans

15/02/24 Data Science

Art of turning raw data into meaningful insights.
→ agriculture, marketing, disease diagnosis.

* What was crop yield when soil condition, weather condition, irrigation pattern.

predictive model and Statistics machine learning.

* drug designing
drug → targets interactions

Tools / Processes of Data Science

1. Data Collection
2. Store Data
3. Pre-process Data
4. Data imputation
5. Data Normalization
6. Describing Data

* Gene Expression Data

	S_1	S_2	S_3	S_4	...	S_{1000}
values	1100	1200	10	7		

Machine Learning Categories

1. Unsupervised learning → Clustering
2. Supervised Learning → classification
3. Reinforcement Learning (Semi Supervised Learning)

* IRIS Data set.

29/02/24

1. Collect
2. Store
3. Process (Preprocessing).
4. Describe
5. Model

Supervised Learning (KNN, SVM, RF, DT) Algorithms
Regression Algorithm

150 flowers
80% Training 120
20% Testing 30

Boston Housing Data Set

No. of rooms, No. length of bedrooms, length of B.R, Width B.R, Height of B.R, Total Carpet Area

Independent variables x

(Y) Target variables Price

$y = f(x)$

Data Processing

1. remove noisy data (outliers
2. Data Imputation (Missing values Imputation)
3. Dimensionality Reduction (Feature Extraction / Feature Selection)

rows are objects (samples)
colum are features.

no. of features (p)
no of objects (s)

* when p >> s (curse of dimensionality)

* Principal component Analysis (PCA).

4. Data Transformation / Data Normalization

1 — 100	0 — 10,000
Min — Max	Standard Scaler (Z-score)
↓	↓
(0, 1)	(−1, 1)

Original value

$$x_i = \frac{x_I - x_{min}}{x_{max} - x_{min}} \quad \Big\} \text{ Min-Max Normalization}$$

F1
x_1 — x_{max}
x_2
x_3
:
x_n — x_{min}

$$x_i' = \frac{x_i - \mu}{\rho} \quad \Big\} \text{ Z-score Normalization}$$

Types of (Data Attributes) / Features

- Numeric - Age, Height, Weight
  - continuous
    - Interval
    - Ratio
  - Discrete
    - ordinal
    - Nominal → Name (identification)
- Categorical
  - Nominal
  - coding scheme

Continuous values

Height → 5.5, 5.51, 5.53 Infinite no. of counts.

5 — 6

Discrete values

Finite no. of counts

Age
1
2
3
:
100

Books
1. Data Mining — P. Tan
2. Data Mining — Han & Kamber

* Incase nominal you can't perform arithmetic operation
* Incase ordinal you can perform subtraction.

5/08/24

Dissimilarity Measures between Data

Similarity

dissim = 1 − sim

Nominal Attributes

$i \underline{\quad} j$ p → no. of features

$d(i,j) = \frac{p-m}{p}$ (m ≤ p)

$d(i,i) = 0$

Symmetric Matrix

$$\begin{array}{c|c} & 1 \quad 2 \quad 3 \quad 4 \\ \hline 1 & \\ 2 & \\ 3 & \\ 4 & \end{array} \begin{bmatrix} 0 & & & \\ 1 & 0 & & \\ 1 & 1 & 0 & \\ 0 & 1 & 1 & 0 \end{bmatrix}$$

d(1,1), d(2,1), d(2,2), d(3,1), d(3,2), d(3,3), d(4,1), d(4,4), d(2,4), d(3,4)

obj-id	test-1 (nominal)	test-2
1	A	
2	B	
3	C	
4	A	

$\frac{1-0}{1}$

$\frac{1-0}{1}$

$\frac{1-0}{1}$

nominal

Binary Attribute

name	gender	fever	cough	test-1	test-2	test-3	test-4
Jack - i	M	Y 1	N 0	P 01	N	N	N
Jim - j	M	Y 1	Y 1	N 0	N	N 0	N
Mary - j	F	Y 1	N 0	P 1	N	P 1	N

$d(\text{jack}, \text{Jim}) = \frac{1+1}{1+1+1} = \frac{2}{3}$ b = 1, c = 1

distinct

Y = P = 1

N = N = 0

$$\begin{array}{c|cc} i \backslash j & 1 & 0 \\ \hline 1 & a\,(1,1) & b \\ 0 & c\,(0,1) & d \end{array}$$

(1,0) (0,0)

∴ a → Y/P

b → N/N

a = 2 → no of columns

$d(\text{Jack}, \text{Mary}) = \frac{b+c}{a+b+c} = \frac{0+1}{2+0+1} = \frac{1}{3}$

$d(\text{Jim}, \text{Mary}) = \frac{1+2}{1+1+2} = \frac{3}{4}$

$d(i,j) = \frac{b+c}{a+b+c+d}$

dissimilarity

Simple Matching Coefficient.

Jaccard Coefficient $= \frac{b+c}{a+b+c}$

Creating List

l = {{1}, {1,2,3}, {"sts"}}

7/03/24

Dissimilarity Measures

Nominal Attribute

Binary Attribute

Interval Scale

Minkowski

$x: (x_1^{A_1}, x_2^{A_2}, x_3^{A_3}, \ldots, x_p^{A_p})$

$y: (y_1, y_2, y_3, \ldots, y_p)$

* Value of Attributes
  - $A_1 \to 0-1$
  - $A_2 \to 1-3$
  - $A_p \to 0-10{,}000$
* All are numeric values

$$d(x,y) = \sqrt[q]{\left(|x_1-y_1|^q + |x_2-y_2|^q + |x_3-y_3|^q + \cdots + |x_p-y_p|^q\right)}$$

when $q=1$

$$d(x,y) = |x_1-y_1| + |x_2-y_2| + \cdots + |x_p-y_p|$$

Manhattan distance

$q=2$

Euclidean distance

Eu. dist $d(x,y) = \sqrt[2]{\left(|x_1-y_1|^2 + |x_2-y_2|^2 + \cdots + |x_p-y_p|^2\right)}$

Ordinal Scale

$\{1 \ldots M_f\}$ → maximum rank

$$z_{if} = \frac{r_{if} - 1}{M_f - 1}$$

$A_5 \to f, \; r^{(5)}$

Days → f

x	Mon	$\to 1 \leftarrow$	$r_x^{(5)}$
y	Wed	$\to 3 \leftarrow$	$r_y^{(5)}$
z	Sun		

$$z_x^{(5)} = \frac{1-1}{7-1} = \frac{0}{6} = 0$$

$$z_x^{(5)} = \frac{7-1}{7-1} = 1$$

Ratio - Scale

$\frac{a}{b}$

$Ae^t, \; Be^t$

ratio $= \frac{A}{B}$

Gene Expression

$\frac{G_1^{S_1}}{G_2^{S_2}}$

S_1 → Normal State

S_2 → Disease state

Ratio conversion to interval

$$\frac{a}{b} \to \log\left(\frac{a}{b}\right) = \log a - \log b$$

Variables of Mixed Types

$A_1, A_2, A_3, \ldots, A_p$

$x_i \quad x_i^{(A_1)} \quad x_i^{(A_2)} \quad x_i^{(A_3)} \ldots x_i^{(A_p)}$

$x_j \quad x_j^{(A_1)} \quad x_j^{(A_2)} \quad x_j^{(A_3)} \ldots x_j^{(A_p)}$

$$d(x_i, x_j) = \frac{d_{A_1}(x_i, x_j) + d_{A_2}(x_i, x_j) + \cdots + d_{A_p}(x_i, x_j)}{p}$$

$$= \frac{\sum_{k=1}^{p} d_{A_k}(x_i, x_j)}{p}$$

weighted dissimilarity between pair objects

$$d(x_i, x_j) = \frac{w_1 * d_{A_1}(x_i, x_j) + w_2 * d_{A_2}(x_i, x_j) + \cdots + w_p * d_{A_p}(x_i, x_j)}{w_1 + w_2 + \cdots + w_p}$$

$$= \frac{\sum_{k=1}^{p} w_k \cdot d_{A_k}(x_i, x_j)}{\sum_{k=1}^{p} w_k}$$

$p >> \cancel{x}$ → Curse of dimensionality

Principle Component Analysis (PCA)

p-dim data set
$\downarrow$
d

$d << p$

μ_x	μ_y	Mean adjusted Data	
x	y	$x-\mu_x$	$y-\mu_y$
x_1	y_1	$x_1-\mu_x$	$y_1-\mu_y$
x_2	y_2	$x_2-\mu_x$	$y_2-\mu_y$
x_3	y_3	$x_3-\mu_x$	$y_3-\mu_y$
$\vdots$	$\vdots$	$\vdots$	$\vdots$
x_m	y_m	$x_m-\mu_x$	$y_m-\mu_y$

PCA

Steps

1. Subtract a mean from each data dimension and find the mean adjusted data.

2. Compute the covariance matrix 'A'.

Covariance matrix for 2 dimensions $= \begin{bmatrix} var(x) & cov(x,y) \\ cov(y,x) & var(y) \end{bmatrix}$

$cov(x,y) = cov(y,x)$

$$var(x) = \frac{\sum(x_i - \mu_x)^2}{m-1}$$

$$cov(x,y) = \frac{\sum(x_i - \mu_x)(y_i - \mu_y)}{m-1}$$

3. Calculate the Eigen values and Eigen vectors of cov Matrix A

$Ax = \lambda x$

$(A - \lambda)x = 0$

$(A - \lambda) = 0$

$\lambda_1, \cdots \lambda_p$

λ_1, λ_2

4. We will arrange the eigen value in descending order

5. From eigen values ~~we will~~ find the eigen vectors.

$Ax = \lambda x$

2D data

λ_1, λ_2

$AX = \lambda_1 X$

$AX = \lambda_2 X$

$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} u_1 \\ u_2 \end{pmatrix}$ λ_1 eigen vector

$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}$ λ_2 eigen vector

6 Transform each eigen vectors into unit length eigenvectors.

$$\begin{pmatrix} u_1' \\ u_2' \end{pmatrix} = \begin{pmatrix} u_1 / \sqrt{u_1^2 + u_2^2} \\ u_2 / \sqrt{u_1^2 + u_2^2} \end{pmatrix}$$

7 Derive the new dataset

$\xrightarrow{} 2\times m$

New dataset = Row Feature Vector * Row Data Adjusted

Feature vector $= \begin{pmatrix} u_1' & v_1' \\ u_2' & v_2' \\ \vdots & \vdots \\ u_p' & v_p' \end{pmatrix} \overset{2D}{=} \begin{pmatrix} u_1' & v_1' \\ u_2' & v_2' \end{pmatrix}$

$p \times p$ dimension

Row Feature Vector = $(\text{Feature Vector})^T \rightarrow 2\times 2$ ~~p×p~~

Row Data Adjusted = $(\text{Mean Adj Data})^T \rightarrow$ no. of elements m, $m\times 2$

$2\times m$

14/03/24 Principal Components Analysis (PCA)

$\bar{x} = 1.81$

$\bar{y} = 1.91$

x	y	$x-\bar{x}$	$y-\bar{y}$
2.5	2.4	0.69	0.49
0.5	0.7	−1.31	−1.21
2.2	2.9	0.39	0.99
1.9	2.2	0.09	0.29
3.1	3	1.29	1.09
2.3	2.7	0.49	0.79
2	1.6	0.19	−0.31
1	1.1	−0.81	−0.81
1.5	1.6	−0.31	−0.31
1.1	0.9	−0.71	−1.01

(Mean adjusted data: columns $x-\bar{x}$, $y-\bar{y}$)

$$A = \begin{pmatrix} \underbrace{cov(x,x)}_{var(x)} & cov(x,y) \\ cov(y,x) & cov(y,y) \end{pmatrix}$$

$$Var(x) = \frac{\sum_i (x_i - \bar{x})^2}{n-1}$$

$$var(y) = \frac{\sum_i (y_i - \bar{y})^2}{n-1}$$

$$cov(x,y) = cov(y,x) = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{n-1}$$

$$= \cancel{\frac{\sum (y_i - y}{9}}$$

$$Var(x) = \frac{5.549}{9} = \cancel{0.6166}\ 0.617$$

$$var(y) = \cancel{\frac{8.70281}{9} = 0.966}\quad \frac{6.449}{9} = 0.716$$

$$cov(x,y) = cov(y,x) = \frac{5.539}{9} = 0.6154$$

$$\Rightarrow A = \begin{pmatrix} 0.617 & 0.6154 \\ 0.6154 & 0.716 \end{pmatrix} \qquad I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$Ax = \lambda x$$

$$\Rightarrow (A - \lambda I)x = 0$$

$$\Rightarrow \begin{pmatrix} 0.617-\lambda & 0.615 \\ 0.615 & 0.716-\lambda \end{pmatrix} = 0$$

$$(0.617-\lambda)(0.716-\lambda)-(0.615)^2=0$$

$$\Rightarrow 0.442-0.617\lambda-0.716\lambda+\lambda^2-0.378=0$$

$$\Rightarrow 0.064-1.33\lambda+\lambda^2=0$$

$$\Rightarrow \frac{1.33\pm\sqrt{(1.33)^2-4\times1\times0.064}}{2\times1}=\lambda$$

$$\Rightarrow \frac{1.33\pm\sqrt{1.777-0.256}}{2}=\lambda$$

$$\Rightarrow \frac{1.33\pm\sqrt{1.521}}{2}$$

$$\Rightarrow \frac{1.33\pm1.233}{2}=\lambda$$

$$\Rightarrow \{\underset{\lambda_1}{1.2815},\ \underset{\lambda_2}{0.485}\}=\lambda$$

$$AX=\lambda X$$

$$AX=\lambda_1 X$$

$$\begin{pmatrix}0.617 & 0.615\\ 0.615 & 0.716\end{pmatrix}\begin{pmatrix}x\\ y\end{pmatrix}=1.283\begin{pmatrix}x\\ y\end{pmatrix}$$

$$0.617x+0.615y=1.283x$$

$$\Rightarrow 1.283x-0.617x=0.615y$$

$$\Rightarrow 0.666x=0.615y$$

$$\Rightarrow x=\frac{0.615}{0.666}y$$

$$\Rightarrow y=\frac{0.666x}{0.615}$$

If we consider $x=1$

$$y=\frac{0.666}{0.615}=1.084$$

$$\begin{pmatrix}x\\ y\end{pmatrix}=\begin{pmatrix}u_1\\ u_2\end{pmatrix}=\begin{pmatrix}1\\ 1.084\end{pmatrix}$$

$$\begin{pmatrix}u_1\\ u_2\end{pmatrix}=\begin{pmatrix}1/\sqrt{1^2+1.084^2}\\ 1.084/\sqrt{1^2+1.084^2}\end{pmatrix}$$

$$\begin{pmatrix}u_1\\ u_2\end{pmatrix}=\begin{pmatrix}0.678\\ 0.735\end{pmatrix}\rightarrow \text{Eigen vector for } \lambda_1$$

$$\begin{pmatrix}0.617 & 0.615\\ 0.615 & 0.716\end{pmatrix}\begin{pmatrix}x\\ y\end{pmatrix}=0.049\begin{pmatrix}x\\ y\end{pmatrix}$$

$$0.617x+0.615y=0.049x.$$

$$\Rightarrow 0.617x-0.049x=-0.615y$$

$$\Rightarrow 0.567x=-0.615y$$

$$\Rightarrow y=\frac{-0.567}{0.615}x$$

If we consider $x=-1$

$$y=\frac{-0.567(-1)}{0.615}=0.922$$

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} -1 \\ 0.922 \end{pmatrix} = \begin{pmatrix} -1 / \sqrt{(-1)^2 + (0.922)^2} \\ 0.922 / \sqrt{(-1)^2 + (0.922)^2} \end{pmatrix}$$

$$\begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} -0.735 \\ 0.678 \end{pmatrix} = \lambda_2 \text{ Eigen Vector}$$

Transformed Data = Row Feature Vector * Row Data Adjusted

$$= (\text{Feature vector})^T * (\text{Mean Adjusted Data})$$

$$= \begin{pmatrix} u_1 & v_1 \\ u_2 & v_2 \end{pmatrix}^T$$

$$= \begin{pmatrix} u_1 & u_2 \\ v_1 & v_2 \end{pmatrix} = \begin{pmatrix} 0.678 & 0.735 \\ -0.735 & 0.678 \end{pmatrix}$$

$$= \underset{2\times 2}{\begin{pmatrix} 0.678 & 0.735 \\ -0735 & 0.678 \end{pmatrix}} * \underset{2\times 10}{\text{Mean Adjusted Data}}$$

$$\text{Transformed Data} = \begin{pmatrix} 0.828 & -1.778 & 0.999 & --- \\ -0.175 & 0.1429 & 0.3844 & --- \end{pmatrix}$$

~~2×10~~

1st principal

2nd principal

2×10

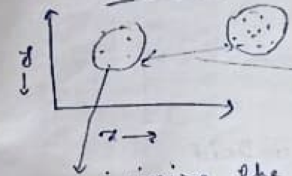
17/03/24

no. of clusters

Clustering ← Unsupervised Learning

K-means

groups → clusters → modules



→ maximize the inter clusters dissimilarity

centroid indicates mean point of clusters

minimize the intra-cluster distance / dissimilarity

1) Partitional Clustering → K-Means, K-Medoid, Partition Around Medoid (PAM), C-Means.

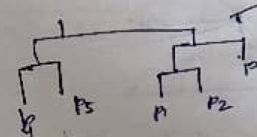
2) Heirerchial clustering (HC) → Agglomerative H.C, Divisive H.C

3) Fuzzy clustering → degree

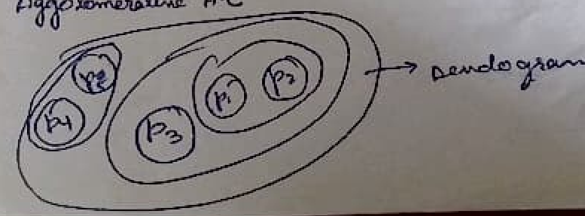
4) Density-based clustering →

→ Dendogram

→ Dendogram



Agglomerative H.C



→ Dendogram

K-Means Clustering
↓
no. of clusters

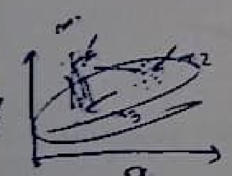
Algorithm

1. Select k points as initial centroid.
2. Repeat
 3. Form k-clusters by assigning an objects to the closest centroid.
4. Recompute the centroid of each cluster.
5. Until the centroids don't change.

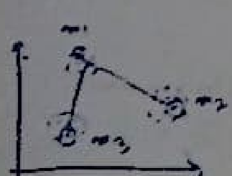
$\sum_{i} \sum_{k=1}^{2} (a_{ik} - \mu_k)$ } try to minimize dissimilarity }

28/03/24

Clustering



2.5 15.7 25.9
Min.



eg

$K = 2$ $C_1 = (1, 1)$ → centroid $C_2 = (2, 1)$ → centroid

x	y	dist (1,1) c_1	dist (2,1) c_2	→ cluster
1	1	0	1	cl 1
2	1	1	0	cl 2 ?
4	3	$\sqrt{13}$	$\sqrt{8}$	cl 2
5	4	5	$\sqrt{18}$	cl 2

$11/3 = 3.67$ $8/3 = 2.67$

Choose minimum value and assigned in that cluster

(i)
$\sqrt{(2-1)^2+(1-1)^2}$
$= \sqrt{1+0}$
$= 1$

(ii)
$\sqrt{(4-1)^2+(3-1)^2}$
$= \sqrt{9+4}$
$= \sqrt{13}$

(iii)
$\sqrt{(4-2)^2+(3-1)^2}$
$= \sqrt{4+4}$
$= \sqrt{8}$

(iv)
$\sqrt{(5-1)^2+(4-1)^2}$
$= \sqrt{16+9}$
$= \sqrt{25}$
$= 5$

(v)
$\sqrt{(5-2)^2+(4-1)^2}$
$= \sqrt{9+9}$
$= \sqrt{18}$

$C_1 = (1,1)$ → remains same

$C_2 = (3.67, 2.67)$ → centroid of the cluster 2.

X	Y	dist(1,1) C_1	dist (3.67, 2.67) C_2	
1	1	0	~~2.75~~ 3.15	cl1
2	1	1	2.36	cl1
4	3	$\sqrt{13}$	0.17	cl2
5	4	5	1.85	cl2

$C_1 = (1.5, 1)$
$C_2 = (4.5, 3.5)$

X	Y	dist(1.5, 1) C2	dist(4.5, 3.5) C2	
1	1	0.5	4.3	Cl1
2	1	0.5	3.53	Cl1
4	3	3.20	0.70	Cl2
5	4	4.61	0.70	Cl2

* when the centroid does not change in the next iteration then we should stop Kmeans process.

$\sqrt{(x^{A_1}-y^{A_1})^2+(x^{A_2}-y^{A_2})^2+(\cdots+(x^{A_n}-y^{A_n})^2}$

Sum of square error (SSE) → try to minimize the. In K means Algorithm

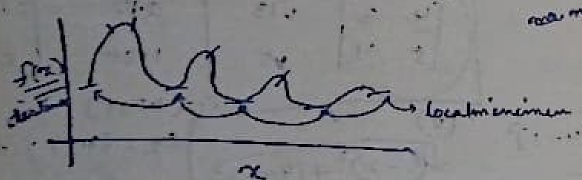
$SSE = \sum_{i=1}^{K} \sum_{x \in i} d^2(x, m_i)$

J

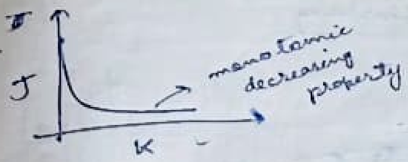
inter cluster distance

centroid for cluster i

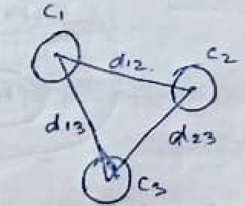
kmeans are only for numeric attributes



K — Medoid
K → Mode
→ non numeric data



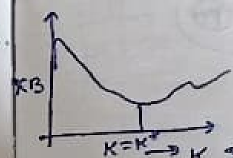
K = m J = 0



Xie – Beni Index

$$XB = \frac{\min \sum_{i=1}^{k} \sum_{x \in C_k} d^2(x, m_i)}{\max\left(n * \min_{i \neq j} (m_i, m_j)\right)}$$

inter cluster distance among all pairs of clusters.



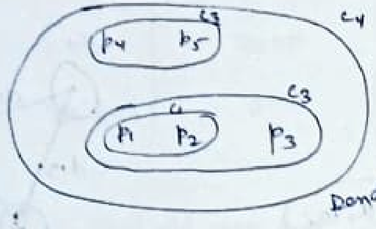
optimal no. of K decided by this

③ K means affected by outliers

16/04/24

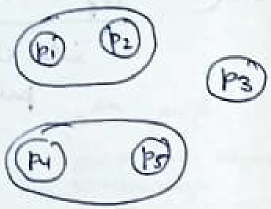
Hierarchical clustering

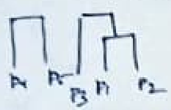
P_1, P_2, P_3, P_4, P_5

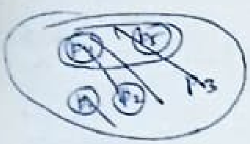


Dendrogram

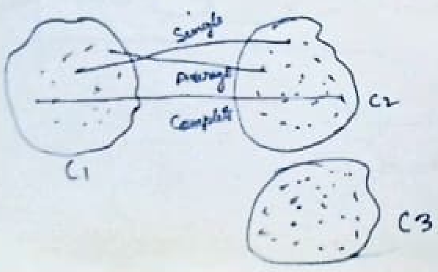
1. Agglomerative H.C
2. Divisive H.C







1. Agglomerative H.C
   - Single Linked A.H.C $\rightarrow \min(d(S_1, S_2), d(S_3, S_2))$
   - Complete " " $\rightarrow \max(d(S_1, S_2), d(S_3, S_2))$
   - Average " " $\rightarrow \text{avg}(d(S_1, S_2), d(S_3, S_2))$



Average Linkage H.C

	A	B	C	D	E
A	0				
B	20	0			
C	60	50	0		
D	100	90	40	0	
E	90	80	50	30	0

$C = \text{avg}(AC, BC)$

$D = \text{avg}(AD, BD)$

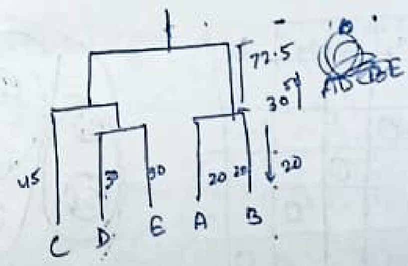
	AB	C	D	E
AB	0			
C	55	0		
D	95	40	0	
E	85	50	30	0

$E = \text{avg}(AE, BE)$

Choose the minimum value coordinate.

	AB	C	DE
AB	0		
C	55	0	
DE	90	45	0

	AB	CDE
AB	0	
CDE	72.5	0



$\frac{55+90}{2} = \frac{145}{2} = 72.5$

Code

```
~~fviz_dend(res.hc,~~
> grp <- cutree (res.hc, k=4)
> head(grp, n=4)
> head(grp, n=50)
> table(grp)
```

18/04/24

Hierarchical Clustering Algorithms

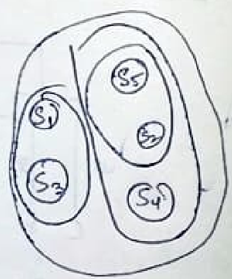
Agglomerative H.C. Algorithm

Steps

1. Compute the distance matrix
2. Let each data point be a cluster
3. Repeat
4. Merge the two closest clusters
5. Update the distance matrix
6. Until only a single cluster remains.

Set of students → $S_1 \ldots S_5$

	S_1	S_2	S_3	S_4	S_5
S_1	0				
S_2		0			
S_3			0		
S_4				0	
S_5					0



SN	x	y
P_1	0.40	0.53
P_2	0.22	0.38
P_3	0.35	0.32
P_4	0.26	0.19
P_5	0.08	0.41
P_6	0.45	0.30

	P_1	P_2	P_3	P_4	P_5	P_6
P_1	0					
P_2	0.23	0				
P_3	0.22	0.14	0			
P_4	0.37	0.19	0.13	0		
P_5	0.34	0.14	0.28	0.23	0	
P_6	0.24	0.24	0.10	0.22	0.39	0

$$d(P_1, P_2) = \sqrt{(0.40 - 0.22)^2 + (0.53 - 0.38)^2} = 0.23$$

$$d(P_1, P_3) = \sqrt{(0.05)^2 + (0.21)^2} = 0.22$$

$$d(P_1, P_4) = \sqrt{(0.14)^2 + (0.34)^2} = 0.37$$

$$d(P_1, P_5) = \sqrt{(0.32)^2 + (0.12)^2} = 0.34$$

$$d(P_1, P_6) = \sqrt{(-0.05)^2 + (0.23)^2} = 0.24$$

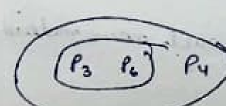
	P_1	P_2	$P_{3,6}$	P_4	P_5
P_1	0				
P_2	0.23	0			
$P_{3,6}$	0.22	0.14	0		
P_4	0.37	0.19	0.13	0	
P_5	0.34	0.14	0.28	0.23	0

$$d(P_1, P_{3,6}) = \min(d(P_1, P_3), d(P_1, P_6))$$

$$d(P_2, P_{3,6}) = \min(d(P_2, P_3), d(P_2, P_6))$$

$$d(P_4, P_{3,6}) = \min(d(P_4, P_3), d(P_4, P_6))$$

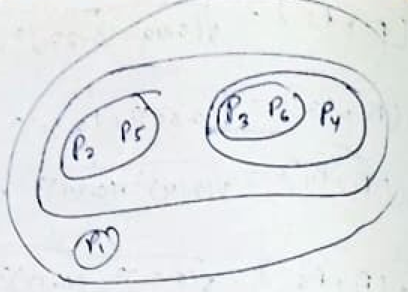
$$d(P_5, P_{3,6}) = \min(d(P_5, P_3), d(P_5, P_6))$$



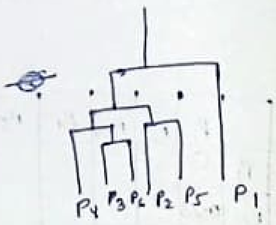
$$d(P_1, P_{3,6,4}) = \min(d(P_1, P_{3,6}), d(P_1, P_4))$$

	P_1	P_2	$P_{3,6,4}$	P_5
P_1	0			
P_2	0.23	0		
$P_{3,6,4}$	0.22	0.14	0	
P_5	0.34	0.14	0.23	0

	P_1	$P_{2,5}$	$P_{3,6,4}$
P_1	0		
$P_{2,5}$	0.23	0	
$P_{3,6,4}$	0.22	0.14	0



	P_1	$P_{2,5,3,6,4}$
P_1	0	
$P_{2,6,4,3,5}$	0.22	0



K- Means → disadvantage

K- medoids Algorithms

Steps

1. Initialize: select K of the 'n' data points as the medoids.
2. Associate each data point to the closest medoid
3. While the cost of the configuration decreases:
4. For each medoid 'm' and for each non-medoid data points 'o':
5. - swap 'm' and 'o' recompute the cost
   - (sum of the distances of points to their medoid)
6. if the Total cost of the configuration increased in the previous step. undo the swap.

Formula for data normalization

$$\bar{x} = \frac{x - \mu}{s}$$

1. load → read.table
2. Summary
3. Scale
4. dist
5. hc = hclust(dist, method = " ")
6. plot (hc)
7. grp = cutree (hc, k = 4 → no. of cls. or h = 0.75)
8. table (grp)
9. head (grp, n=50)

K-medoids

PAM

Partition Around Medoids

(x_1, y_1)
(x_2, y_2)

$d = |x_1 - x_2| + |y_1 - y_2|$

Cluster		X	Y	$M_1(3,4)$	$M_2(6,4)$	Cl Assign K=2
C_1	x_1	2	6	3	6	C_1
C_1	x_2	3	4	0	3	C_1
C_1	x_3	3	8	4	7	C_1
C_2	x_4	4	2	3	4	C_1
C_2	x_5	6	2	5	2	C_2
C_2	x_6	6	4	3	0	C_2
	Sum of Error SE			10	2	

$x_2 = (3,4) \leftarrow C_1$

$x_6 = (6,4) \leftarrow C_2$

(TSE) Total Sum of Error = 10 + 2 = 12.

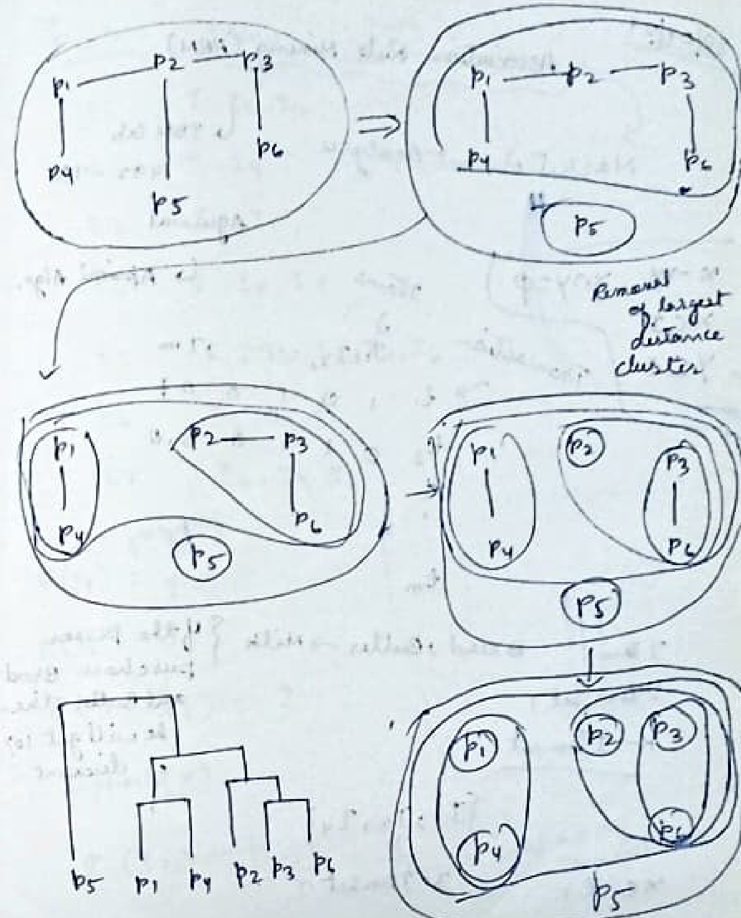
M_1 (3,4)	M_2 (6,2)	Cl Assign	M_2' (4,2)	
3	8	C_1	6	C_1
0	5	C_1	3	C_1
4	9	C_1	7	C_1
3	2	C_2	0	C_2
5	0	C_2	2	C_2
3	2	C_2	4	C_1
SE: 7	4 ①		② SE = 2	

② 7+3=10

① TSE = 7 + 4 = 11

② TSE = 10 + 2 = 12

Divisive Hierarchical Clustering



Algorithm

① Compute a MST for the proximity graph.

② Repeat

③ Create a new cluster by removing the link corresponding to the largest distance (smallest similarity).

(4) Until only singleton clusters remains.

30/04/24

Association Rule Mining (ARM)

Market Basket Analysis

IBM lab 1992-93

Agarwal

Apriori Algo.

$X \to Y$ $X \cap Y = \phi$

$X \subset I$

$Y \subset I$

Items

$I_1, I_2, \ldots, I_m$

Transaction

t_1 1 0 1 0 .. 0 1

t_2 0 1 1 0 ... 0 0

⋮

t_m

Item : Bread, Butter → Milk { If the person purchase Bread and Butter then he will get 10% discount. }

Item set ,

K- Item set

$\{I_1, I_3, I_4\}$

3-Itemset

$X \in t_1$

$X \subseteq I$

~~Support count~~ (SC).

1. Support - count (δ)

Transaction

TID	Items
t_1	I_1, I_2, I_3
t_2	I_1, I_4
t_3	I_4, I_5
t_4	I_1, I_2, I_4
t_5	I_1, I_2, I_3, I_4, I_6
t_6	I_2, I_3, I_6
t_7	I_2, I_3, I_6

frequency

$\delta(I_1) = 4$

$\delta(I_5) = 1$

$\delta(I_1, I_2) = 3$

2. Support (S)

$S(I_1) = \frac{|I_1|}{n} = \frac{4}{7}$ → Total no. of Transactions

$S(I_1, I_2) = \frac{|\{I_1, I_2\}|}{m} = \frac{3}{7}$

3. Support of an association rule

$X \to Y$

$S(X \to Y) = \frac{|X \cup Y|}{m}$

$S(\{I_1, I_2\} \to \{I_3\}) = \frac{|\{I_1, I_2, I_3\}|}{m} = \frac{2}{7}$

4. Confidence of an association rule (c).

$$c(X \to Y) = \frac{|X \cup Y|}{|X|}$$

5. Frequent Itemset / Large Itemset

$S(\{I_1, I_2\}) = \frac{3}{7} \approx 43.n\%$

min-sup = 30%

$X \to Y$ ← consequent

antecedent

$\{I_1, I_2\}$

$$C(I_1 \to I_2) = \frac{|I_1, I_2|}{|I_1|} = \frac{3}{4} = 75\%.$$

$$C(I_2 \to I_1) = \frac{|I_2, I_1|}{I_2} = \frac{3}{5} = 60\%.$$

min-sup = 30%

min-conf = 70%

1/5/24

Association Rule Mining (ARM)

Frequent itemsets, support, confidence.

Apriori Algo → for generating ARM.
↳ Agarwal, 1993

1. Generating all possible frequent itemsets.
2. Generating association Rules from frequent itemsets.

Apriori Algo (Simplified) $\xrightarrow{\text{based on}}$ Downward closure Property

- Any subset of Frequent itemsets are also frequent.
- conversely any superset of an infrequent itemset are also infrequent

purchased / Frequent $\langle I_1, I_3, I_4 \rangle$

Infrequent $\times \langle I_4, I_7 \rangle$

not purchased

$\times \langle I_7, I_9, I_8 \rangle$

$\times \langle I_4, I_6, I_7 \rangle$

Apriori Algo. is a

- iterative algo.
- First of all find all 1-itemsets ~~~~ frequent item sets then find all 2 item frequent itemsets then find all k item frequent itemsets.
- In each iteration k(say) → only consider itemsets that ~~~~ contains same (k-1) frequent itemsets.

Algorithm

- Find frequent itemsets of size $K=1$; F_1

Main step:

2. from $K=2$
3. C_K = Candidate itemsets of size K: these itemsets of size K that could be frequent, given F_{K-1}.
4. F_K = those itemsets that are actually frequent $F_K \subseteq C_K$.

(Need to scan the database again).

Output: $F = F_1 \cup F_2 \cup \ldots \cup F_K$

Support: $S(x \to y) = \frac{|x||y|}{m}$

$\text{confidence}(x \to y) = \frac{|x \cup y|}{|x|}$

Frequent itemsets

itemsets whose support is greater than min support (sup)

$\boxed{\text{sup (itemsets)} \geq \text{min-sup}}$

no. of itemsets

$I_1, I_2 \ldots I_{10}$

$F_1: \langle I_1\rangle, \langle I_2\rangle, \langle I_5\rangle, \langle I_7\rangle \longrightarrow$ Frequent item sets.

$C_2 = \langle I_1, I_2\rangle$
$\langle I_1, I_5\rangle$
$\langle I_1, I_7\rangle$
$\langle I_2, I_5\rangle$
$\langle I_5, I_7\rangle$
$\langle I_2, I_7\rangle$

F_2

* Combine the itemset which has different items in second position

$F_2: \langle I_1, I_2\rangle, \langle I_1, I_7\rangle, \langle I_2, I_5\rangle, \langle I_5, I_7\rangle$
$\langle I_2, I_7\rangle$

$C_3 = \cancel{\langle I_1, I_2, I_7\rangle}$ → There is nothing common in it then we will not combine it.

Frequent

$\langle I_1, I_5\rangle$ is infrequent

$\langle I_1, I_2, I_7\rangle, \langle I_1, I_2, I_5\rangle, \langle I_2, I_5, I_7\rangle, \langle I_1, I_5, I_7\rangle$

$\langle I_1, I_5\rangle$

$\langle I_2, I_5, I_7\rangle \leftarrow$ ~~Fre~~ Frequent

$F_3 = C_3 = \langle I_1, I_2, I_7\rangle, \langle I_2, I_5, I_7\rangle$

$C_4 = \langle I_1, I_2, I_5, I_7\rangle \leftarrow \langle I_1, I_2, I_5\rangle$ → is not present.

$F_4 = \phi$

Output
$\therefore F = F_1 \cup F_2 \cup F_3 \cup F_4 \Rightarrow \phi$

eg

T ID	Items
t_1	1,3,4
t_2	2,3,5
t_3	1,2,3,5
t_4	2,5

$C_1 \to F_1 \to C_2 \to F_2 \to C_3 \to F_3 \to C_4 \to F_4 \to \ldots$

C_1

<1> : 2

<2> : 3

<3> : 3

<4> : 1 ✗ min support

<5> : 3

F_1 : <1>, <2>, <3>, <5>

C_2 : <1,2>, <1,3>, <1,5>, <2,3>, <2,5>, <3,5>
 : 1 : 2 : 1 : 2 : 3 : 2

F_2 : <1,3>, <2,3>, <2,5>, <3,5>

C_3 : <1,2,3>, <2,3,5>, <1,3,5>
 ✗ : 1 : 2 ✗ : 1

F_3 : <2,3,5>

11/12/24

FP-Tree growth Algorithm

Advantages:-

- Scan the transaction database only twice.

	descending order of support count
t_1 : I_1, I_2, I_5	t_1 : I_2, I_1, I_5
t_2 : I_2, I_4	t_2 : I_2, I_4
t_3 : I_2, I_3	t_3 : I_2, I_3
t_4 : I_1, I_2, I_4	t_4 : I_2, I_1, I_4
t_5 : I_1, I_3	t_5 : I_1, I_3
t_6 : I_2, I_3, I_6	t_6 : I_2, I_3
t_7 : I_1, I_3	t_7 : I_1, I_3
t_8 : I_1, I_2, I_3, I_5	t_8 : I_2, I_1, I_3, I_5
t_9 : I_1, I_2, I_3, I_7	t_9 : I_2, I_1, I_3

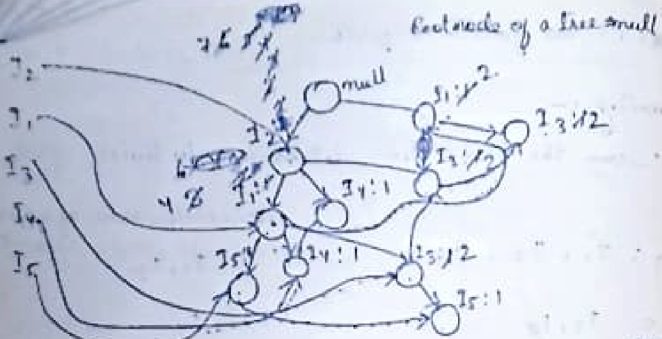
* Min-sup : 2

* $S(I) = |I|$

$I_1 = 6$
$I_2 = 7$
$I_3 = 6$
$I_4 = 2$
$I_5 = 2$
✗ $I_6 = 1$

$I_7 = 1$

$I_2 = 7$
$I_1 = 6$
$I_3 = 6$
$I_4 = 2$
$I_5 = 2$



FP- tree

Items	conditional Pattern base	conditional FP-Tree	Frequent Patterns
I_5	$(I_2, I_1 ; 1)$, $(I_2, I_1, I_3 ; 1)$	$\langle I_2:2, I_1:2 \rangle$	$(I_2, I_5 : 2)$, $(I_1, I_5 : 2)$, $(I_1, I_2, I_5 : 2)$
I_4	$(I_2, I_1 : 1)$, $(I_2 : 1)$	$\langle I_2 : 2 \rangle$	$\{I_2, I_4 : 2\}$
I_3	$(I_2 : 2)$, $(I_2, I_1 ; 2)$, $(I_1 : 2)$	$\langle I_2 : 4, I_1 : 2 ; I_1 : 2 \rangle$	$\langle I_2, I_3 : 4)$, $(I_1, I_3 : 2)$, $(I_1, I_2, I_3 : 2)$
I_1	$I_2 : 4$	$\langle I_2 : 4 \rangle$	$(I_2, I_1 : 4)$
I_2	—	—	—

directly from rootnode

$(I_2, I_1, I_5 : 2)$

$X \rightarrow Y \rightarrow$ consequent

antecedent

$(I_2, I_1) \rightarrow I_5$

$s(X \rightarrow Y) = \frac{|X \cup Y|}{|m|} = \frac{2}{9} \simeq 0.022$

min-sup = 20%   min-conf = 50%

$conf(X \rightarrow Y) = \frac{|X \cup Y|}{|X|} = \frac{2}{4} = 0.5$

$m \rightarrow$ no. of items in transaction

$(2^m - 2) \rightarrow$ no. of database created

$(I_2, I_5 : 2)$

$I_2 \rightarrow I_5 \rightarrow$ $sup(I_2 \rightarrow I_5) = \frac{2}{9} \simeq 0.222$

$I_5 \rightarrow I_2$ $conf(I_2 \rightarrow I_5) = \frac{2}{7} \simeq 0.285 = 0.29$

$sup(I_5 \rightarrow I_2) = \frac{2}{9} \simeq 0.222$

$conf(I_5 \rightarrow I_2) = \frac{2}{2} = 1.0 > 0.5$

valid association rule

14/11/24

$$\text{Lift}(T\to C) \overset{X\to Y}{=} \frac{m \times sc(X\cup Y)}{sc(X)\ sc(Y)}$$

Lift/ $(X\to Y)$

support count

$$\text{Lift}(T\to C) = \frac{100\times 15}{20\times 90} < 1 \to \text{Invalid}$$

$$\text{Lift}(\bar{T}\to C) = \frac{100\times 75}{80\times 90} > 1 \to \text{valid}$$

valid association rule

$$C(\bar{T}\to C) = \frac{75}{80} > .90$$

$X\to Y$

Support, confidence

Supervised
- → Classification (Machine Learning)
- → Regression

Height	Weight	Age	Gender (class label)
–	–	–	–
–	–	–	–
–	–	–	–

$f(H, W, A) \to G$

Classification

Training dataset
- → Training set
- → Test set

5000 students (H, W, A, G)

80:20

Tr → 4000 students

Tst → 1000 student (H, W, A)

↓

G_p → Predicted gender

Tr

$M(H, W, A) \to G$

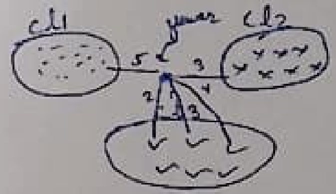
Test

$M(H, W, A) \to G_p$ (Predicted) G_A (Known)

compare

KNN (K-nearest neighbours) for class level.

Classification



$K = 5$

$\{3, 3, 2, 3, 1\}$

$\{3, 2, 3, , 2, 1\}$

$d_1 \quad d_2 \quad d_3 \quad d_4 \quad d_5$

K is increase

$(d_1 + d_3) < (d_2 + d_3)$ 4

KNN Algorithm

1. Store all the input data in the training set.
2. For each object in the test set
3. Search for K nearest neighbours ~~to the~~ (objects) to the input objects using any distance metrics. (Euclidean)
4. For classification, compute the confidence for each classes as C_i / K, where C_i is the no. of objects among the K-nearest neighbours belonging to class label 'i'.
5. The classification for the input object is the class with the highest confidence.

$$\frac{C_1}{K} = \frac{1}{5}$$

$$\frac{C_2}{K} = \frac{1}{5}$$

$$\frac{C_3}{K} = \frac{3}{5}$$

eg

class label

Sl	CGPA	Asgn	Project	Result
1	9.2	8.5	8	Pass
2	8	8.0	7	Pass
3	8.5	8.1	8	Pass
4	6	4.5	5	Fail
5.	6.5	5.0	4	Fail
6.	8.2	7.2	7	Pass
7	5.8	3.8	5	Fail 1
8.	8.9	4.1	9	Pass

$(6.1, 4.0, 5) \xrightarrow{?}$ Result $K = 3$

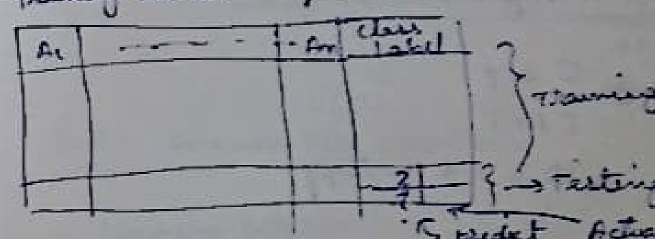
9.2 8.5 8

$$\sqrt{(9.2 - 6.1)^2 + (8.5 - 4.0)^2 + (8 - 5)^2}$$

$= 6.0728$ $K = 5$

Decision Tree Classification

Training Dataset → Attributes / Features



$f(A_1, A_2, \ldots, A_n) \rightarrow$ Class Label

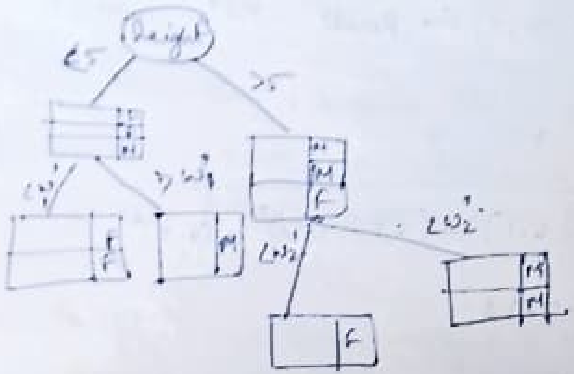
Decision Tree

○ → Root Node

◇ → Internal / Decision

□ → Leaf Node

Height	Weight	Gender
h_1	w_1	M
h_2	w_2	F
h_3	w_3	M
h_4	w_4	M
h_5	w_5	F
h_6	w_6	F



Hunt's Algorithm

- C 4.5
- CART
- ID3 → Entropy

Entropy: degree of uncertainty associated with a variable.

coin (H, T)

Dice (1, 2, ----, 6)

higher uncertainty → higher Entropy

lower " → lesser Entropy

$$\text{Entropy} = -\sum_{i=1}^{m} P_i \cdot \log P_i$$

eg 10 → 6 passed and 4 failed.

$$\text{Entropy} = -\left(\frac{6}{10} \cdot \log_2\left(\frac{6}{10}\right) + \frac{4}{10} \log_2\left(\frac{4}{10}\right)\right)$$

P_1 P_2

(1, 0)

$$-\left(\frac{100}{100} \log_2 \frac{100}{100} + \frac{0}{100} \log \frac{0}{100}\right)$$

$= 0 + 0$

$= 0$

eg 100 → 100, 0

Homogeneous data → entropy value is lower.

ID3 Decision Tree Algorithm

- T → Training Dataset
- A → $A_1, A_2 ... A_m$ ← Attributes
- m → no. of classes
- P_i → probability that a data instance belongs to class C_i.
  $i = (1 --- , m)$

data

$$P_i = \frac{|A_i|}{|T|}$$

P_i → Total no. of data instances belongs to class C_i

$$\text{Entropy-Info}(T) = -\sum_{i=1}^{m} P_i \log_2 P_i \quad \text{---(1)}$$

$$\text{Entropy-Info}(T,A) = \sum_{i=1}^{v} \frac{|A_i|}{|T|} \times \text{Entropy-Info}(A_i)$$

$10 \longrightarrow 3 \leftarrow$ tall

$\frac{3}{10}$ 2 → short $\frac{2}{10}$

Entropy

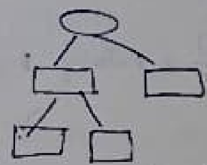
Information Gain (A)

$$= \text{Entropy-Info}(T) - \text{Entropy-Info}(A) \quad \text{---(3)}$$

higher Info Gain ↓

ID3 Algorithm

1. Compute Entropy-Info (T) ← eq(1) for the whole training dataset
2. Compute Entropy-Info (A) and Info Gain (A) eq(2) and eq(3) for each attribute A in the training dataset
3. Choose the attribute for which entropy is minimum and therefore info gain is maximum.
   → best split attribute.
4. The best split attribute is placed as the root node.
5. The root node is branched into subtrees with each subtree as an outcome of the test condition of the root node attribute.
   The training dataset is also split into subtrees.
6. Recursively apply the same procedure for the subset of the training dataset with remaining attributes until a leaf node is derived or no more training instances are available in the subset.



Eg 1

Class Labels

CGPA	Interactiveness	Practical knowledge	Communication Skill	Job Offer
≥9	yes	very good	Good	yes
≥8	No(N)	G(Good)	Moderate (M)	Yes(Y)
≥9	N	Average	Poor	N
<8	N	A	G	N
≥8	Y	G	M	Y

≥ 9	Y	A	N	N
< 8	Y	G	P	N
≥ 9	N	V.G	G	Y
≥ 8	Y	G	G	Y
≥ 8	Y	A	G	Y

$$\text{Entropy-Info}(T) = -\sum_{i=1}^{m} P_i \cdot \log_2 P_i \qquad \left(\log_2 a = \frac{\log_{10} a}{\log_{10} 2} \right)$$

$$\text{Entropy-Info}(\text{Job offer}) = -\left(\frac{7}{10} \cdot \log_2 \frac{7}{10} + \frac{3}{10} \log_2 \frac{3}{10} \right)$$

$$= -(-0.3599 + (-0.5208))$$

$$= 0.8807$$

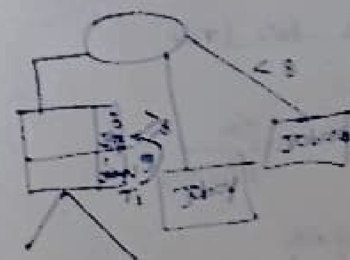
$$\boxed{\text{Entropy-Info}(T, A) = \sum_{i=1}^{v} \frac{|A_i|}{|T|} \times \text{Entropy-Info}(A_i)}$$

Entropy-Info (T, CGPA)

CGPA	Job=Yes	Job=No	Total	Entropy
≥ 9	3	1	4	
≥ 8	4	0	4	0
< 8	0	2	2	0

Entropy-Info (T, CGPA)

=



Entropy-info (T_1) = 0.8108

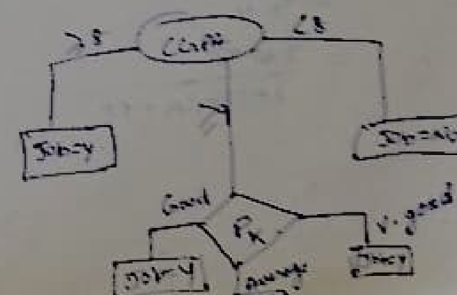
Entropy-info $(T_1$, Intern) = 0.4997

⇒ I.G. (Intern) = 0.8108 − 0.4997

= 0.311

I. Gain $(T_1$, P.S) = 0.8108

I. Gain $(T_1$, CS) = 0.8108



4) True Negative Rate (TNR)

$$= \frac{d}{c+d} = \frac{TN}{FP+TN}$$

↖ specificity

5) Precision

p

$$: \frac{a}{a+c} = \frac{TP}{TP+FP}$$

6) F_1-Score (F-measure)

$$= \frac{2 * \overset{\text{recall}}{r} * \overset{\text{precision}}{p}}{r+p}$$

$$= \frac{2*a}{2a+b+c}$$

$$= \frac{2*TP}{2*TP+FN+FP}$$

11.10

TP 35	FP 5
FN 10	TN 30

accuracy, recall, precision, F_1-score.